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Module 2 Write Up

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Introduction

Module 2 focused on computer forensics and explored the principles, methods, tools, and even some real examples of computer forensics. The major purpose of computer forensics is to understand or recreate something that happened on a system. The types of cases that computer forensics affect people across society and will almost certainly become even more important going into the future, as computers become even more common. Examples of computer forensics cases are fraud, IP theft, data breaches, and child pornography. The process of computer forensics involves gather evidence, Investigating, Analyzing Data, and Reporting.

Evidence

In any kind of forensics evidence is the crucial pieces that are used to understand the case in question. In computer forensics the evidence can seem ethereal because it often exists in a digital format that live in computers and must be accessed using a computer. Despite this difficult there is still a large amount of evidence that can be gathered from computers, and there are many methods to investigate to find this information. It is important to initially create copies of any logical stores on any devices, networking devices, removeable media to preserve the state of the scene. Preservation of evidence in computer forensics is extremely important and sometimes difficult. It is important to create cryptographic hashes of files and then work with copies of the original data. These hashes can be used to verify that whenever you are working with data it is the same as the original. Unlike traditional forensics we have the luxury of easily making copies, but the integrity of computer evidence is fragile, if all copies are lost and the original is modified even slightly it will likely become useless.

Investigating

The Major types of evidence to focus on in a computer are the memory, the logical drive, and any network traffic. These pieces of evidence can be investigated for specific information in several ways. Memory can be dumped and examined using tools such as AccessData FTK and give real time information on things the system is doing. For me reaching the information in FTK was difficult but I was able to pick out a few strings that were understandable. The logical drive of the computer system also has a huge amount of information and can be investigated using many different tools such as the command line, windows explorer, and tools like AccessData FTK. Network data can also be reviewed using many windows tools, the command line, and various public tools.

Analyzing Data

To get specific information from evidence it must be analyzed. In computer forensics we use various tools to extract specific information from the before mentioned evidence, things that can be obtained are a long list such as

- Running Process
- Device drivers
- Open files
- Open network ports
- Keystrokes
- Emails and attachments
- Usernames / Passwords
- Personal files
- System Events and Logs
- And much more depending on the nature of the case.
- USB Fingerprint.

This information can help create a timeline or narrative of what happened on a system and can provide extremely useful information for building a case against an individual.

Windows Registry

The windows registry is an extremely store that contains records virtually everything done in windows, obviously this will contain tons of useful forensic information. Tools like RegMon and Regedit can be used to monitor or check values in registry, which can then gather evidence. I had accessed the registry plenty of times in my life but I will admit I didn't really understand the structure. The registry is composed of five hives, the five HKEY or Handle to a key. Apparently only two are real, the HKEYUSERS and HKEY_LOCAL_MACHINE, while the rest are shortcuts, again very new information to me. Inside the registry we can find lost files using the master file table to create a timeline of what happened on the machine. In the example we used volatility to examine a \$MFT file and see files that had been created, this would be crucial in any kind of forensic case involving the system.

Lab 1

Lab 1 is about understanding and reverse engineering a GDB bomb, however technically I have not finished this lab at the time of writing this. From the lab, this file is an executable which requires specific input from the user but has no external indications as to what the input should be. The challenge here will be to determine the inputs and “defuse” each of the phases. My first thought about this lab is there seems like there are only so many ways to really interact with a computer so the domain of what inputs I must try shouldn’t be that large. However, then I reconsidered and questioned what does it really mean to be an input, a trigger could be something fair more subtle like interacting with a file a certain way or required an action be performed at certain time intervals. While I can’t speak the specifics that I will have tried in the lab I look forward to attempting the challenge. The registry also contains urls typed into internet explorers and even a log of all USB devices that have been plugged into the system.

Conclusion

Computer forensics is a quickly rising area of criminal investigation. The evidence, tools, and methods in computer forensics require special understanding and can be difficult to interpret but can yield critical information. Especially in computer focused cases computer forensics can indeed be the only real evidence available. In this module we explored several tools to examine windows, personal files, the windows register, and even data that was in memory using a memory dump. Often even when using the tool we had to have some other computer understanding to interpret what we were seeing. Often while working through these examples I thought about how difficult it must be for none computer educated individuals who are involved in these kind of cases.

Appendix

This write-up follows the MLA standard for expository essays.